

FINAL EXAM

ECO403–INTRODUCTION TO CONTRACT THEORY & AUCTIONS

MAY 2025

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Final Exam Instructions and Rules

Please, read this carefully before attempting anything!

The final examination consists of two parts:

- A **take-home list of problems**, to be completed in groups.
- A **group oral examination**, during which *each student will be asked individual questions* to assess their personal understanding and contribution.

Collaboration and Submission

- Students must work in **groups of up to 5 members**. A deviation of +1 is allowed *only with prior written consent* from the instructor.
- Only **one written report** is required per group (unless attempting *only Section 1*, in which case **no written submission** is necessary).
- Each group must send **one email** to the instructor stating the **full list of group members**.
- **Inter-group consultation is permitted**, but **each group must write their own solutions independently**.
- Students are **strictly prohibited** from discussing this exam with **anyone outside the class**, or from sharing oral exam content with others after their interview.
- **All written submissions must be typed**, including all mathematical expressions.
 - It is strongly recommended to use $\text{\LaTeX}$ (e.g., Overleaf) or Microsoft Word with an equation editor.
 - **Handwritten mathematics embedded as images will not be accepted.**
 - This is a formal report and should be treated accordingly.
 - If required, the submission must consist of a single PDF file sent to my email.
 - Your submission must be received **before 20:00 on Wednesday, May 7, 2025, through Moodle. Only one submission per team.**
 - Your written solutions should be **careful, rigorous, and succinct**. There are **penalties for unnecessarily long or poorly structured arguments**. Knowing how to express a mathematical argument clearly and concisely is part of what is being evaluated.

Use of Resources

- You may consult:
 - Your own class notes and printouts of the slides.
 - Any book listed as a reference in the syllabus.
 - Any other book or online source you deem appropriate.
- **All references must be properly cited and acknowledged.**
- You may use **any computer algebra system** (e.g., Mathematica, WolframAlpha, Maple) for final calculations. However, you must provide a **complete logical argument**, omitting only routine algebra when appropriate.

Use of Generative AI Tools

- Use of **Generative AI tools** (e.g., ChatGPT, Claude, Copilot, DeepSeek, etc.) is permitted **only up to 25%** of your written submission.
- You are fully responsible for the **correctness and understanding** of any AI-generated content.
- During the oral exam, I will assess whether each student understands the material. If it becomes clear that AI did the thinking, **this constitutes academic dishonesty**.

Integrity and Consequences

- **Uploading the exam paper to any internet repository or database is strictly prohibited.**
- **Any violation** of these rules will be penalized under the Academic Integrity Policy. Even a **Category A violation** may result in **failing the course**, at my discretion and that of the Academic Discipline Committee.

Instructions and Grading Criteria

Let $\mathcal{G} = \{F, D, D+, C-, C, C+, B-, B, B+, A-, A, A+\}$. We partition $\mathcal{G}$ as follows:

$$G_0 = \{F, D, D+\}$$

$$G_1 = \{C-, C, C+\}$$

$$G_2 = \{B-, B, B+\}$$

$$G_3 = \{A-, A\}$$

$$G_4 = \{A+\}$$

The exam has **four sections**. You should attempt problems based on the grade for the final exam that you aim to achieve.

Target Grade in $G_0 \cup G_1$

- Prepare for **only Section 1**.
- **No written submission** is required.
- In your group email indicating your team members, please also let me know your intention to attempt only this part.
- Grades based on oral exam:
 - **Weak performance** $\rightarrow G_0$
 - **Good performance** $\rightarrow G_1$

Target Grade in $G_0 \cup G_1 \cup G_2$

- Prepare for **Section 1** (no written submission).
- Submit written solutions to:
 - At least one problem from Section 3 (Bayesian Games and Beliefs),
 - At least one problem from Section 4 (Auctions).
- Grades based on written + oral:
 - **Very weak** $\rightarrow G_0$
 - **Weak** $\rightarrow G_1$
 - **Average, OK** $\rightarrow G_2$

Target Grade in $G_0 \cup G_1 \cup G_2 \cup G_3$

- Prepare for **Section 1** (no written submission).
- Submit written solutions to:
 - At least one problem from Section 2 (Knowledge),
 - At least one problem from Section 3 (Bayesian Games and Beliefs),
 - At least two problems from Section 4 (Auctions).
- Grades based on written + oral:
 - **Very weak** $\rightarrow G_0$
 - **Weak** $\rightarrow G_1$
 - **Average** $\rightarrow G_2$
 - **Good or very good** $\rightarrow G_3$

Target Grade in G_4

The grade **A+** does not appear on transcripts, but it is my personal distinction for students who have **gone above and beyond** and earned my **respect and admiration**.

- You must attempt **all problems in all four sections**.
- All previous grading rules apply.
- Performance must be **exceptional**.

Students earning an A+ will receive:

- An **automatic offer for an outstanding recommendation letter**,
- A chance to become a **Teaching Assistant** for this course,
- The opportunity to be a **Research Assistant**,
- Opportunities to be **supervised in an MA thesis or capstone project**,
- And: **Your name will be etched in my memory—and in future exams and problem sets—as a symbol of excellence.**

Problems in the next page. Good luck!



1 Auction Vignettes: Overall Evaluation of Course Content

This task contains three real-world-inspired vignettes about auctions in Kazakhstan. Your task is to read and reflect deeply on each situation. During the oral examination, I will ask questions derived from these vignettes. You may be asked to:

- Discuss possible auction models and information structures.
- Present informal or formal arguments.
- Evaluate strategic behavior of agents.
- Identify connections to main course topics and the concepts we covered within, such as **Knowledge & Beliefs**, **Bayesian Games**, **SIPV Auctions**, **Revenue Equivalence**, **Reserve Prices & Entry Fees**, and **Interdependent Value Auctions**.

Think of this as presenting a *business case*—because I know just how much you adore those. Be ready to discuss each vignette at three levels: **informally** (as if you’re briefing a non-economist director who just wants “the takeaway”), **semi-formally** (as if advising a competent economist who actually understands the mechanics), and a bit more **formally**—say, if you were trying to impress a sharp professor...like me.

You are not required to submit anything written for this task. But be ready to think precisely, speak clearly, and—if asked—write on the whiteboard.

1.1 The Privatization Puzzle—Kazakhstan, 1994

In September 1994, Kazakhstan auctioned off over 1,600 state-owned enterprises as part of its rapid privatization program. The format was an **open outcry ascending auction**. Prices rose dramatically — in some cases, sevenfold. The result caused widespread concern: some bidders claimed they had no reliable data on firms and regretted their bids. Subsequently, the government reverted to **sealed-bid auctions**, citing instability and overvaluation.

1.2 Refining the Bid—The Stratus Oil Auction (2022–2024)

In 2022, Mangystau Oil Refining (MOR) acquired the bankrupt Stratus Oil refinery in Turkestan through a **public auction**. Many bidders exited early, citing a lack of transparency and concerns over environmental liabilities. MOR submitted a strong bid and was declared the winner. However, in 2024, the Kazakh court invalidated the deal, stating the auction had failed to produce a competitive or legitimate outcome and lacked procedural safeguards.

1.3 Wind Power for Sale—Renewable Energy Auctions (2018–2022)

To attract investment in green energy, Kazakhstan held a series of **reverse auctions** for wind and solar power. Bidders submitted sealed bids indicating the price at which they were willing to sell electricity to the grid. Winning bids were often 30–40% below previous tariffs. Some winning firms later struggled to deliver on their commitments, raising questions about “overly aggressive bidding” and whether auction incentives had induced inefficiencies or unsustainable outcomes.

2 Knowledge

2.1 Fun with the standard model of Knowledge

Consider¹ a knowledge space $(N, \Omega, (\mathcal{P}_i)_{i \in N})$, where N and Ω are finite sets. Let K_i be the knowledge operator that we studied in class, i.e., for an event $A \in 2^\Omega$, $K_i(A) = \{\omega \in \Omega : P_i(\omega) \subseteq A\}$.

2.1.1. Give an example of a knowledge space and an event A for which a player does not know A nor A^c at any state of the world.

¹This problem was kindly suggested to me by our TA.

2.1.2. Characterize the events A for which a player does not know A nor A^c at any state of the world.

2.1.3. Characterize the information partitions $\mathcal{P}_i$ for which there always exists an event A such that player i does not know A nor A^c in any of the states of the world.

2.2 Even more fun with non-standard models of Knowledge

Let N be a finite set of players and Ω a finite set of states of the world. Consider an arbitrary² knowledge operator

$$K_i : 2^\Omega \rightarrow 2^\Omega,$$

mapping events (subsets of Ω) to events.

Consider the following axioms that can be imposed on the knowledge operator K_i :

T (Truth Axiom) $K_i(A) \subseteq A$ “if i knows A , then A must be true;”

K (Distributivity) $K_i(A) \cap K_i(B) = K_i(A \cap B)$ “knowing A and B is equivalent to knowing A and knowing B ;”

N (Necessitation) $K_i(\Omega) = \Omega$ “player i always knows anything that is true in all states of the world;”

M (Monotonicity) If $A \subseteq B$, then $K_i(A) \subseteq K_i(B)$ “If A implies B then knowledge of A implies knowledge of B ;”

PI (Positive Introspection) $K_i(A) \subseteq K_i(K_i(A))$ “if i knows A , then she knows that she knows it;”

NI (Negative Introspection) $\Omega \setminus K_i(A) \subseteq K_i(\Omega \setminus K_i(A))$ “if i does not know A , then she knows that she does not know it.”

2.2.1. For any knowledge operator K_i , show that **K** implies **M**.

2.2.2. Show that if K_i satisfies **K** and **N**, then it generates a possibility correspondence $P_i : \Omega \rightarrow 2^\Omega$ defined by

$$P_i(\omega) = \bigcap \{E \subseteq \Omega \mid \omega \in K_i(E)\}.$$

That is, show that P is a well-defined function mapping each state ω to a subset of Ω .

Now suppose that $P'_i : \Omega \rightarrow 2^\Omega$ is a possibility correspondence. Define its associated knowledge operator as

$$K'_i(A) = \{\omega \in \Omega \mid P'_i(\omega) \subseteq A\}.$$

Show that K'_i satisfies **K** and **N**.

2.2.3. Suppose K_i satisfies **K** and **N** as in the previous item. Furthermore, suppose it also satisfies **T**, **PI**, and **NI**. Show that the possibility correspondence

$$P_i(\omega) = \bigcap \{E \subseteq \Omega \mid \omega \in K_i(E)\}$$

is a *partition correspondence*; that is, the collection $\{P_i(\omega)\}_{\omega \in \Omega}$ forms a partition of Ω .

You do not need to prove the converse (that if we start with a partition correspondence and define the corresponding knowledge operator, it satisfies **K**, **N**, **T**, **PI**, and **NI**)—we already established this in class.

²The knowledge operator K_i described here does not necessarily arise from a possibility correspondence or a partition correspondence. Recall that a *possibility correspondence* is a function $P_i : \Omega \rightarrow 2^\Omega$, where $P_i(\omega) \subseteq \Omega$ represents the set of states that agent i considers possible at state ω . A *partition correspondence* is a possibility correspondence that generates a partition of Ω ; that is, the collection $\{P_i(\omega)\}_{\omega \in \Omega}$ forms a partition of Ω .

3 Beliefs & Bayesian Games

3.1 Contribution towards a public good...Once again!

Consider the following public good production game. There are n players who may decide whether or not to contribute to the provision of a public good. The public good is produced only if at least k players choose to contribute, where $1 \leq k \leq n$. Contribution is a binary decision: either you contribute or you don't, and the amount contributed is fixed.

Each contributor incurs a cost $c > 0$, which is the same for all players and publicly known. On the other hand, each player's valuation of the public good, v_i , is private information: individual i knows her own valuation v_i , but not those of the other players. However, it is common knowledge that all valuations lie within the interval $[\underline{v}, \bar{v}]$, where $0 \leq \underline{v} < c < \bar{v}$. Each player believes that the probability that any given individual's valuation is at most v is given by the distribution function $F(v)$, independently across individuals, where F is a continuous cumulative distribution function. Furthermore, F admits a probability density function f .

The game proceeds as follows: all n individuals simultaneously submit envelopes. In each envelope, a player may include their fixed contribution c , or leave it empty (i.e., contribute nothing). If fewer than k players choose to contribute, the public good is not produced, all contributions are returned, and the payoff for each player is zero. If at least k players contribute, then the public good is produced, contributions are accepted, and the payoffs are as follows:

$$u_i(a_i, a_{-i}, v_i) = \begin{cases} v_i - c & \text{if } a_i = 1 \text{ and } \sum_{j=1}^n a_j \geq k, \\ v_i & \text{if } a_i = 0 \text{ and } \sum_{j=1}^n a_j \geq k, \\ 0 & \text{if } \sum_{j=1}^n a_j < k. \end{cases}$$

- 3.1.1. Provide the belief space description of the problem above and describe formally the set of pure strategies of each player. Does the payoff structure correspond to a private values setting, an interdependent values setting, or neither? Justify briefly.
- 3.1.2. Find conditions under which there exists a *pure strategy Bayesian Nash equilibrium* in which exactly k players contribute, and all other players do not contribute.³ That is, identify conditions under which an *asymmetric equilibrium* exists such that: each contributing player has a sufficiently high valuation and chooses to contribute, all remaining players do not contribute regardless of their valuation, and the number of contributors exactly matches the contribution threshold k .
Comment on whether multiple such asymmetric equilibria can exist and explain why.
- 3.1.3. Focus on the case in which the contribution threshold is $k = 1$. Find a symmetric pure strategy Bayesian Nash equilibrium in which each player contributes if and only if her valuation exceeds some critical threshold v^* .
- 3.1.4. At the equilibrium you identified in the previous item, what is the overall probability that the public good is provided? How does this probability depend on n ? What happens as $n \rightarrow \infty$?
- 3.1.5. Generalize your results about the symmetric equilibrium to the case of a general contribution threshold k .

3.2 How to Lose Your Degree Because of Cheating: A Game Theoretic Analysis of NU's Academic Disciplinary Committee

A student in ECON403/503 has been suspected of academic dishonesty. Dr. A, bound by both moral duty and academic protocol to make life difficult for such students, reports the case to the appropriate academic disciplinary committee. According to Nazarbayev University's academic integrity policy, the case is then referred to a jury consisting of two faculty members from the School of Sciences and Humanities (SSH), who

³If you cannot solve this for general k , assume $k = 1$.

must evaluate the situation and reach a verdict. The committee requires a *unanimous verdict*: the student is convicted only if both jurors vote to convict. Otherwise, the student is acquitted.

Each juror receives a private signal about the case. This signal is intended to reflect underlying evidence of the student's guilt or innocence. Let the state of nature be denoted $\theta \in \{G, I\}$, where G indicates that the student is *guilty* and I indicates that the student is *innocent*. The prior probability of guilt is denoted $\mu = \Pr(\theta = G)$, where $\mu \in (0, 1)$.

Each juror privately observes a signal $s_i \in \{A, E\}$, where A denotes an *accusatory* signal and E denotes an *exculpatory* signal. The joint distribution of signals (s_1, s_2) depends on the true state θ , and is parameterized by three quantities. The parameter $p \in (1/2, 1)$ captures the likelihood of receiving an accusatory signal when the student is guilty, while $q \in (1/2, 1)$ captures the likelihood of receiving an exculpatory signal when the student is innocent. The parameter $\xi \in [\underline{\xi}, \bar{\xi}]$ governs the degree of correlation between the jurors' signals, with the lower and upper bounds defined by the requirement that all joint probabilities lie in the unit interval.

The joint distribution of signals conditional on the state of nature is given below.

$\theta = G$	$s_2 = A$	$s_2 = E$	$\theta = I$	$s_2 = A$	$s_2 = E$
$s_1 = A$	$p^2 + \xi$	$p(1-p) - \xi$	$s_1 = A$	$(1-q)^2 + \xi$	$(1-q)q - \xi$
$s_1 = E$	$p(1-p) - \xi$	$(1-p)^2 + \xi$	$s_1 = E$	$(1-q)q - \xi$	$q^2 + \xi$

After receiving their respective signals, each juror must take an action $a_i \in \{0, 1\}$, where action 1 denotes a vote to *convict* and action 0 a vote to *acquit*. A conviction occurs if and only if both jurors vote to convict.

Jurors receive payoffs based on the true state and the final verdict. If the student is guilty and convicted, or innocent and acquitted, the juror receives a payoff of 0. If the student is *innocent and convicted*, the juror incurs a cost of $-z$. If the student is *guilty and acquitted*, the juror incurs a cost of $-(1-z)$, where $z \in (0, 1)$.

- 3.2.1. Define a state of the world to be the triple $\omega = (\theta, s_1, s_2)$, where $\theta \in \{G, I\}$ is the true state of nature, and $s_1, s_2 \in \{A, E\}$ are the private signals received by the two jurors. Let Ω denote the set of all such states. Provide a complete description of the belief space by explicitly writing the prior distribution over Ω , and defining the signalling functions that map each state $\omega \in \Omega$ to the signal observed by each juror.
- 3.2.2. Complete the description of the Bayesian game by writing down the payoff function of each juror as a function of the state $\omega = (\theta, s_1, s_2)$ and the action profile. Describe formally the set of pure strategies of each player. Does the game exhibit private values, interdependent values, or neither? Justify your answer briefly.
- 3.2.3. Let r denote the posterior probability that the student is guilty, given the juror's information. Compute the expected payoff to a juror when the defendant is acquitted, and the expected payoff when the defendant is convicted. Compare these expressions and interpret the role of the parameter z in determining the juror's preferences over the two outcomes.
- 3.2.4. Consider the conditional distributions of signals given the state $\theta \in \{G, I\}$. Compute the values of $\underline{\xi}$ and $\bar{\xi}$ that ensure the resulting joint distributions are valid probability distributions; that is, all entries lie in the interval $[0, 1]$.
- 3.2.5. For the remainder of the problem, unless otherwise specified, assume $\xi = 0$; that is, the signals are conditionally independent given the state. Assume further that the signals are ordered as $A > E$. Compute the joint distribution over signals, $\Pr(s_1, s_2)$, by marginalizing over the state θ . Based on this distribution and the assumed ordering, determine whether the signals s_1 and s_2 are affiliated. Justify your answer.
- 3.2.6. Consider the symmetric strategy profile in which each juror follows a monotonically increasing strategy based on their signal: voting to acquit when receiving signal E , and voting to convict when receiving signal A . Derive the conditions on the parameter z under which this strategy profile constitutes a Bayesian Nash Equilibrium in pure strategies.

- 3.2.7. Recall that a *type I error* occurs when an innocent defendant is convicted, while a *type II error* occurs when a guilty defendant is acquitted. Under the symmetric equilibrium described in the previous item—where each juror votes to acquit upon receiving signal E and to convict upon receiving signal A —compute the joint probability mass function over the state of nature and the equilibrium action profile; that is, compute the probability of each tuple (ω, a_1^*, a_2^*) , where a_i^* denotes the equilibrium strategy of juror i . Using this joint pmf, compute the probability that the committee makes a type I error and the probability that it makes a type II error.
- 3.2.8. Now suppose that $\xi \in [\underline{\xi}, \bar{\xi}] \setminus \{0\}$; that is, the signals are no longer conditionally independent given the state. Repeat the previous two items: (i) derive the conditions on z under which the symmetric monotone strategy profile remains a Bayesian Nash Equilibrium, and (ii) compute the probability that the committee makes a type I or type II error under this equilibrium. Compare these results to those obtained under $\xi = 0$. What role does the correlation parameter ξ play in shaping equilibrium incentives and institutional accuracy?

4 Auctions

4.1 SIPV model: k -th price auctions

There are $n \geq k \geq 2$ bidders competing for a single indivisible object. Bidder i 's valuation is denoted $V_i \sim U[0, 1]$, independently and identically distributed across bidders. Each bidder observes her own realization v_i , but not those of others. Assume the remaining standard assumptions from the *symmetric independent private values (SIPV)* environment.

The seller conducts a *k -th price auction*, in which each bidder submits a sealed bid. The object is awarded to the highest bidder, who pays the k -th highest bid among all those submitted. Bids are unrestricted, except that they must be strictly positive real numbers. All bidders are risk-neutral and aim to maximize their expected payoff. Ties, if any, are broken uniformly at random.

- 4.1.1. Propose at least two distinct methods that could be used to solve for a symmetric, monotonically increasing, and differentiable Bayesian Nash equilibrium bidding strategy in the k -th price auction described above. Your answer should clearly state the idea behind each method, but you are not required to carry out the derivation.
- 4.1.2. Using an appropriate version of the Revenue Equivalence Theorem (RET), solve for a symmetric, strictly increasing, and differentiable equilibrium bidding strategy in the k -th price auction. You should apply the ex-ante version of the RET, which equates expected payments from the point of view of the auctioneer across all auctions that implement the same allocation rule. To guide your solution, assume that the equilibrium bidding strategy takes the linear form $\beta_k^*(v) = \alpha_{n,k}v$, where the constant $\alpha_{n,k}$ depends on the number of bidders n and the format of the auction (that is, the value of k). Use the RET to derive the value of this constant. Then, you should prove that the bidding function you derived is indeed an equilibrium.
- 4.1.3. Analyze the bidding function you derived in the previous question. Are any bidders bidding above their own valuation in equilibrium? Would any bidder ever end up paying more than her own valuation? Justify your answer.
- 4.1.4. Let R_k denote the random variable representing the revenue earned by the auctioneer in the k -th price auction. Derive the distribution function of R_k explicitly. Your answer should be fully worked out and expressed in terms of the number of bidders n , the auction format k , and any relevant order statistics.
- 4.1.5. Compute the variance of R_k , the revenue in the k -th price auction, and show that it increases in k . Then, prove that R_k second-order stochastically dominates R_{k+1} for all $k = 1, \dots, n-1$. Be precise in your definitions and arguments.

4.2 Common Value First-Price Auction: Exploring the Winner's Curse and Regret

Consider a sealed-bid *first-price auction* with two bidders, where the *common value* of the object is given by $V = X_1 + X_2$, with X_1 and X_2 being independent and identically distributed random variables on the interval $[0, 1]$, with cumulative distribution function F that is commonly known among the bidders.

Each bidder $i \in \{1, 2\}$ privately observes their own signal X_i , and knows the distribution of the other bidder's signal.

Assume the distribution function is $F(x) = x^\theta$, for $x \in [0, 1]$, where $\theta > 0$, with corresponding density $f(x) = \theta x^{\theta-1}$.

Each bidder submits a bid based on their private signal. The highest bidder wins the object and pays their own bid.

- 4.2.1. Compute the symmetric, monotonically increasing, and differentiable equilibrium bidding function of the first-price auction. Analyze the bidding function with respect to the parameter θ . Are there any cases in which a bidder bids above her own signal?
- 4.2.2. Compute $E[V \mid X_1 = x, Y_1 < x]$, where Y_1 is the highest order statistic from the sample excluding player 1. Use this to argue that, no matter the value of θ , player 1 is never a victim of the winner's curse at the interim stage when using the symmetric equilibrium strategy.
- 4.2.3. Suppose that both players follow the equilibrium strategies, and that after the auction concludes, player 1 is the winner and the bid of player 2 is also announced. For each value of θ , characterize the region of regret for player 1: the pairs (x, y) such that player 1 wins but regrets doing so because they overpaid. Compute both the interim probability of regret and the ex-ante probability of regret for player 1. Does this contradict the assertion of the previous item?

4.3 Symmetric Interdependent Value Auctions: Two Interesting Formats

This problem follows the standard assumptions from the symmetric interdependent value auctions we covered in class. There is a single object to be auctioned and there are n bidders. Each bidder i receives a real-valued signal X_i with support $[0, \omega]$. The value of the object to bidder i is $V_i = u(X_i, X_{-i})$, where u is non-negative, continuous, strictly increasing in all of its arguments, and symmetric with respect to its last $n - 1$ arguments. It is also assumed that $E[V_i] < \infty$ for all i .

The random variables $X_1, \dots, X_n$ have a joint density $f(X_1, \dots, X_n)$ that is symmetric and affiliated. Denote $Y_1 = \max_{j \neq 1} X_j$. Let $g(y \mid x)$ denote the conditional density of Y_1 given that $X_1 = x$, and let $G(y \mid x)$ denote the corresponding cumulative distribution function. Let

$$\lambda(y \mid x) = \frac{g(y \mid x)}{1 - G(y \mid x)}$$

denote the *hazard rate* of the distribution of Y_1 conditional on $X_1 = x$. Define, as usual,

$$v(x, y) = E[V_1 \mid X_1 = x, Y_1 = y],$$

the expected value of the object for bidder 1, conditional on her own signal and the maximum signal among the other bidders.

Take the following as facts:

Fact 1 X_1 and Y_1 are affiliated.

Fact 2 $G(y \mid x)/g(y \mid x)$ is non-increasing in x .

Fact 3 $\lambda(y \mid x)$ is non-increasing in x .

Fact 4 $G(y \mid x)$ is non-increasing in x .

Fact 5 $v(x, y)$ is strictly increasing in both arguments.

We will now examine two different auction formats under this environment.

In the first format, a *war of attrition* auction, each bidder submits a sealed bid b_i , and the payoffs are given by:

$$W_i^{\text{war}} = \begin{cases} V_i - \max_{j \neq i} b_j & \text{if } b_i > \max_{j \neq i} b_j, \\ -b_i & \text{if } b_i < \max_{j \neq i} b_j, \\ \frac{1}{\#\{k: b_k = b_i\}} (V_i - b_i) & \text{if } b_i = \max_{j \neq i} b_j, \end{cases}$$

where ties are broken by awarding the object to each of the highest bidders with equal probability.

In the second format, we consider an *all-pay* auction. Each bidder again submits a sealed bid b_i , but now the payment is made regardless of winning. The payoffs are:

$$W_i^{\text{all}} = \begin{cases} V_i - b_i & \text{if } b_i > \max_{j \neq i} b_j, \\ -b_i & \text{if } b_i < \max_{j \neq i} b_j, \\ \frac{1}{\#\{k: b_k = b_i\}} (V_i - b_i) & \text{if } b_i = \max_{j \neq i} b_j, \end{cases}$$

As before, in the case of a tie, the object is awarded to each of the highest bidders with equal probability.

- 4.3.1. Derive the symmetric equilibrium bidding functions for both auction formats described above. Let $\beta(x)$ denote the bidding function in the war of attrition, and let $\alpha(x)$ denote the bidding function in the all-pay auction. Assume that both β and α are symmetric, strictly increasing, and differentiable functions. Provide a heuristic derivation of each function, carefully justifying all your steps based on the auction rules and payoff structures.
- 4.3.2. Derive any additional conditions required to ensure that the functions $\beta(x)$ and $\alpha(x)$ are indeed monotonically increasing. Explicitly state and assume these conditions for the remainder of your analysis.
- 4.3.3. Verify that the bidding functions $\beta(x)$ and $\alpha(x)$ you derived constitute symmetric Bayesian Nash equilibria in their respective auctions.
- 4.3.4. Consider the equilibrium strategies derived in the previous questions. Show that, for each signal x , the interim expected payment of a bidder with signal x is greater than the interim expected payment in the second-price auction when the format is a war of attrition. Similarly, show that the interim expected payment in the all-pay auction is greater than the interim expected payment in the first-price auction. Clearly state any assumptions you make about equilibrium bidding strategies in the first-price and second-price auctions, and justify each comparison carefully.